Test Series No.

03

NEET Success Kit [2026-27]

DURATION: 195 Minutes | **M. MARKS: 720**

DATE: 10/06/2026

ANSWER KEY

1.	(2)	46.	(1)	91.	(2)	136.	(2)
2.	(1)	47.	(3)	92.	(3)	137.	(2)
3.	(2)	48.	(4)	93.	(2)	138.	(3)
4.	(2)	49.	(3)	94.	(2)	139.	(4)
5.	(3)	50.	(1)	95.	(2)	140.	(4)
6.	(3)	51.	(2)	96.	(2)	141.	(3)
7.	(2)	52.	(1)	97.	(3)	142.	(4)
8.	(4)	53.	(2)	98.	(2)	143.	(4)
9.	(4)	54.	(1)	99.	(2)	144.	(4)
10.	(2)	55.	(3)	100.	(2)	145.	(2)
11.	(2)	56.	(2)	101.	(2)	146.	(4)
12.	(3)	57.	(4)	102.	(1)	147.	(4)
13.	(1)	58.	(3)	103.	(4)	148.	(4)
14.	(2)	59.	(2)	104.	(2)	149.	(4)
15.	(1)	60.	(3)	105.	(2)	150.	(4)
16.	(2)	61.	(2)	106.	(2)	151.	(4)
17.	(1)	62.	(4)	107.	(1)	152.	(1)
18.	(1)	63.	(1)	108.	(1)	153.	(1)
19.	(1)	64.	(2)	109.	(2)	154.	(1)
20.	(4)	65.	(3)	110.	(2)	155.	(2)
21.	(4)	66.	(1)	111.	(4)	156.	(4)
22.	(3)	67.	(1)	112.	(4)	157.	(3)
23.	(2)	68.	(1)	113.	(3)	158.	(4)
24.	(4)	69.	(1)	114.	(4)	159.	(4)
25.	(3)	70.	(2)	115.	(3)	160.	(1)
26.	(2)	71.	(1)	116.	(1)	161.	(1)
27.	(3)	72.	(1)	117.	(4)	162.	(4)
28.	(1)	73.	(2)	118.	(4)	163.	(2)
29.	(1)	74.	(1)	119.	(1)	164.	(1)
30.	(4)	75.	(4)	120.	(2)	165.	(3)
31.	(2)	76.	(1)	121.	(4)	166.	(3)
32.	(2)	77.	(1)	122.	(2)	167.	(4)
33.	(1)	78.	(1)	123.	(4)	168.	(2)
34.	(2)	79.	(3)	124.	(4)	169.	(2)
35.	(1)	80.	(1)	125.	(4)	170.	(2)
36.	(1)	81.	(4)	126.	(2)	171.	(3)
37.	(1)	82.	(2)	127.	(1)	172.	(4)
38.	(1)	83.	(4)	128.	(3)	173.	(1)
39.	(2)	84.	(3)	129.	(4)	174.	(2)
40.	(3)	85.	(3)	130.	(2)	175.	(3)
41.	(4)	86.	(1)	131.	(2)	176.	(4)
42.	(2)	87.	(4)	132.	(3)	177.	(4)
43.	(1)	88.	(1)	133.	(3)	178.	(3)
44.	(3)	89.	(1)	134.	(4)	179.	(1)
45.	(2)	90.	(2)	135.	(1)	180.	(1)

[Contd...

1. (2)

We know $e = -\frac{Ldi}{dt}$

for $0 \le t \le \frac{T}{4}, \frac{di}{dt} = 0$

so, $e = 0$

for $\frac{T}{4} < t \le \frac{3T}{4}$, $\frac{di}{dt}$ = negative and constant

so, $e \rightarrow$ positive and constant

And for $\frac{3T}{4} < t \le T, \frac{di}{dt} = 0$

$\therefore\ e = 0$

(NCERT Ed. 2025-26, 12th, Page No. 158)

2. (1)

$a = -A\omega^2 \cos\omega t$

$v = \int a dt$

$= -A\omega^2 \int \cos\omega t\, dt$

$= -A\omega^2 \frac{\sin\omega t}{\omega} = -A\omega \sin\omega t + c$

(NCERT Ed. 2025-26, 11th, Page No. 267)

3. (2)

Increase in forward current leads to increase in flow of charge carries across junction. Hence decrease in width of depletion layer.

(NCERT Ed. 2025-26, 12th, Page No. 335)

4. (2)

Magnitude: $B = \frac{E}{c} = \frac{12}{3\times10^8} = 4\times10^{-8}$ T.

Direction: $\vec{E}\times\vec{B}$ points in the direction of propagation ($+z$).

Since $\vec{E}$ is $+x$ ($\hat{i}$), $\vec{B}$ must be along $+y$ ($\hat{j}$) because $\hat{i}\times\hat{j} = \hat{k}$.

(NCERT Ed. 2025-26, 12th, Page No. 208)

5. (3)

EF due to infinite charged sheet is uniform & independent of distance.

(NCERT Ed. 2025-26, 12th, Page No. 34)

6. (3)

Speed of all EM radiations is same in vacuum.

(NCERT Ed. 2025-26, 12th, Page No. 211)

7. (2)

The inward flux, $\phi = \vec{E}.\vec{A}$

$= (7\hat{j}).\pi(1)^2(-\hat{j})$ due to flat surface of cone

$= -7\pi$ Vm

(NCERT Ed. 2025-26, 12th, Page No. 23)

8. (4)

Power factor for an ideal choke coil as well as for a purely capacitance circuit is zero.

(NCERT Ed. 2025-26, 12th, Page No. 183)

9. (4)

During free fall the weighing machine will show zero reading.

(NCERT Ed. 2025-26, 11th, Page No. 52)

10. (2)

When the body is dropped, its velocity is zero. Then it speeds up under the effect of gravity in downward direction. When it rebounds, its velocity starts decreasing in upward direction & again becomes zero when it reaches the highest point.

(NCERT Ed. 2025-26, 11th, Page No. 20)

11. (2)

For different frequencies, we have difference stopping potentials and as intensity is fixed, they will have same saturation current.

(NCERT Ed. 2025-26, 12th, Page No. 278)

12. (3)

Houses made of concrete roofs get very hot during summer days because thermal conductivity of concrete (though much smaller than that of a metal) is still not small enough.

(NCERT Ed. 2025-26, 11th, Page No. 215)

13. (1)

$v_e = \sqrt{\frac{2GM}{R}} = \sqrt{\frac{2GMR^2}{R^3}}$

For same mass density $\left(\frac{M}{R^3}\right)$

$v_e \propto R$

$\therefore$ From the given planet

$v'_e = 5v$

(NCERT Ed. 2025-26, 11th, Page No. 136)

[Contd...

14. (2)

Fringe width $\beta = \frac{\lambda D}{d}$

In the liquid of refractive index μ, wavelength decreases $\left(\lambda \propto \frac{1}{\mu}\right)$. So β also decreases to $\frac{1}{\mu}$ times.

$\therefore\ \beta' = \frac{\beta}{1.6} = 0.625\beta$

(NCERT Ed. 2025-26, 12th, Page No. 266)

15. (1)

$Z = 1$ for hydrogen and $Z = 3$ for lithium

We know, $v_n = \frac{Z}{n} v_0$

$(v_5)_H = \frac{1}{5} v_0$ and $(v_2)_{Li^{++}} = \frac{3}{2} v_0$

$(v_5)_H = v$ (given)

$\therefore\ (v_2)_{Li^{++}} = \frac{3}{2}(5v) = \frac{15}{2} v = 7.5v$

(NCERT Ed. 2025-26, 12th, Page No. 299)

16. (2)

Area $A = \pi r^2 \Rightarrow \frac{\Delta A}{A} = 2\frac{\Delta r}{r}$

The lateral/radial strain is $\frac{\Delta r}{r} = \frac{-2.4\%}{2} = -1.2\%$

Poisson's ratio is $\sigma = -\frac{\Delta r / r}{\Delta L / L}$

Longitudinal strain $\frac{\Delta L}{L} = -\frac{-1.2\%}{0.4} = 3.0\%$

(NCERT Ed. 2025-26, 11th, Page No. 174)

17. (1)

Using Bernoulli's equation for a horizontal pipe:

$P_1 + \frac{1}{2}\rho v_1^2 = P_2 + \frac{1}{2}\rho v_2^2$

$P + \frac{1}{2}\rho v^2 = P_2 + \frac{1}{2}\rho (2v)^2$

$P + \frac{1}{2}\rho v^2 = P_2 + 2\rho v^2$

$P_2 = P + \frac{1}{2}\rho v^2 - 2\rho v^2 = P - \frac{3}{2}\rho v^2.$

(NCERT Ed. 2025-26, 11th, Page No. 188)

18. (1)

Clockwise *P-V* curve implies positive work done by the gas & anticlockwise implies negative work.

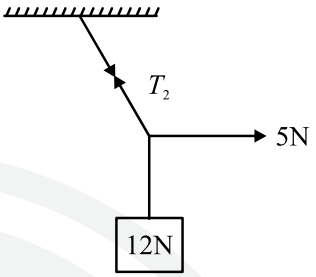
So, net work is positive.

(NCERT Ed. 2025-26, 11th, Page No. 235)

19. (1)

$T_1 = 12$ N

and T_2 can be found by equilibrium equation.



$T_2 = \sqrt{5^2 + 12^2} = 13$ N

$\therefore\ T_1 : T_2 = 12 : 13$

(NCERT Ed. 2025-26, 11th, Page No. 59)

20. (4)

For equilibrium, $mg = Bil$

$m = \frac{4.9 \times 1 \times 5}{9.8} = 2.5$ kg

(NCERT Ed. 2025-26, 12th, Page No. 110)

21. (4)

The emf induced across centre & the circumference of the disc $= \frac{1}{2} B\omega R^2$.

So, e_{AB} = zero as potential on the boundary is constant.

(NCERT Ed. 2025-26, 12th, Page No. 164)

22. (3)

Let $i_1 = 3i$ across $5R$

Now, $i_2 = \frac{1}{3} i_1 = i$

$i_3 = \frac{2}{3} i_1 = 2i$

$\therefore\ i_1 : i_2 : i_3 = 3 : 1 : 2$

(NCERT Ed. 2025-26, 12th, Page No. 100)

[Contd...

23. (2)

All fringes have same width and all bright fringes have same intensity.

(NCERT Ed. 2025-26, 12th, Page No. 263)

24. (4)

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

The output is 0 only when both inputs are 1.
This corresponds to NAND gate.

(NCERT Ed. 2025-26, 12th, Page No. 341)

25. (3)

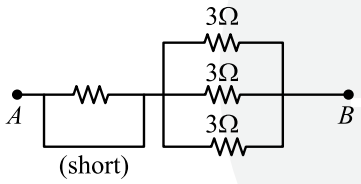
While connected to battery, the potential difference across capacitor plates remains constant.

And, $C = \frac{\varepsilon_0 A}{d}$ So, $C \propto \frac{1}{d}$

(NCERT Ed. 2025-26, 12th, Page No. 74)

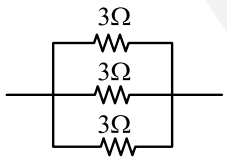
26. (2)

(I)



$$R_{AB} = \left(\frac{1}{3}+\frac{1}{3}+\frac{1}{3}\right)^{-1} = 1\Omega$$

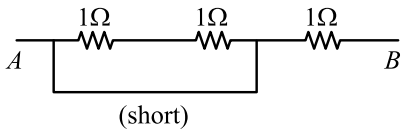
(II)



$R_{AB} = 1\Omega$

(III) $R_{AB} = 1+1 = 2\Omega$

(IV)

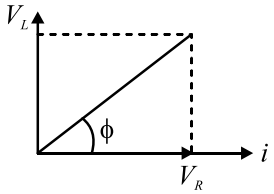


$R_{AB} = 1\Omega$

(NCERT Ed. 2025-26, 12th, Page No. 100)

27. (3)

In series LR AC circuit, voltage leads current by phase difference ϕ.



(NCERT Ed. 2025-26, 12th, Page No. 187)

28. (1)

The magnetic flux in the coil first increases as the particle approaches the coil & then decreases as it recedes away. The direction of induced current opposes the cause. Hence it is first anticlockwise then clockwise.

(NCERT Ed. 2025-26, 12th, Page No. 160)

29. (1)

We know, $v_w = \frac{\omega}{k}$ and $v_{max} = A\omega$

$v_w = \frac{1}{2} v_{max}$ (given)

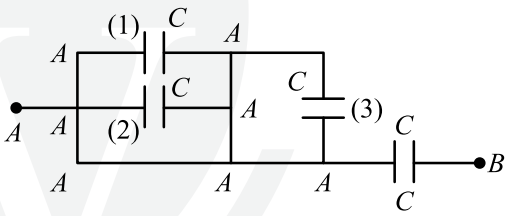
$\frac{\omega}{k} = \frac{1}{2} A\omega$

$\frac{\lambda}{2\pi} = \frac{1}{2} A$

$\therefore \lambda = \pi A$

(NCERT Ed. 2025-26, 11th, Page No. 284)

30. (4)



capacitors (1), (2) & (3) are shorted.

$\therefore C_{AB} = C$

(NCERT Ed. 2025-26, 12th, Page No. 73)

31. (2)

Cutting the magnet perpendicular to its length into two equal parts results in each piece having a magnetic moment $M' = \frac{M}{2}$.

When the two pieces are stacked with "like poles together," the total magnetic moment of the system becomes $M_{total} = M' + M' = \frac{M}{2} + \frac{M}{2} = M$.

Since the total magnetic moment M remains the same as the original magnet and the magnetic field B is constant, the potential energy $U = -MB\cos\theta$ remains U.

(NCERT Ed. 2025-26, 12th, Page No. 139)

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32. (2)

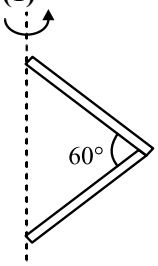
Work done = ΔKE (Work energy theorem)

$= \frac{1}{2}m\left((0)^2 - (20)^2\right)$

$= \frac{1}{2} \times 5(0 - 400) = -1000\ \text{J}$

(NCERT Ed. 2025-26, 11th, Page No. 76)

33. (1)



$I = \frac{2m(l\cos\theta)^2}{3} = \frac{2ml^2\cos^2 30°}{3}$

$= \frac{2}{3}(2)(1)^2\left(\frac{\sqrt{3}}{2}\right)^2 = \frac{4\times 3}{3\times 4} = 1\,\text{kg m}^2$

(NCERT Ed. 2025-26, 11th, Page No. 116)

34. (2)

Energy = Flux × Time × Area

$= 10 \times 2 \times 60 \times 50 = 6 \times 10^4$ J

(NCERT Ed. 2025-26, 12th, Page No. 284)

35. (1)

$H = A\sigma T^4$

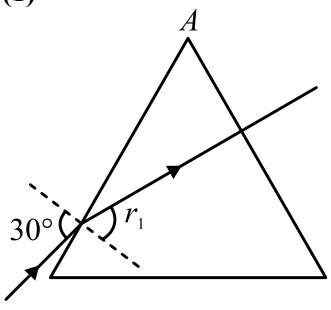
Per unit area $(H \propto T^4)$ Let new rate = R'

$\Rightarrow \frac{R'}{R} = \left(\frac{2T}{T}\right)^4$

$R' = 16R$

(NCERT Ed. 2025-26, 11th, Page No. 219)

36. (1)



Here, $i = 30$, $\mu = 1.5$, $r_2 = 0°$

$\mu = \frac{\sin i}{\sin r_1}$

Also $A = r_1 + r_2$

$A = r_1$

$\therefore\ 1.5 = \frac{\sin 30°}{\sin A}$

$\sin A = \frac{1}{2} \times \frac{1}{1.5} = 0.33$

(NCERT Ed. 2025-26, 12th, Page No. 240)

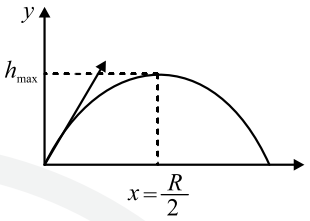
37. (1)

Given eqn. of trajectory $= y = x - \frac{x^2}{40} = x\left(1 - \frac{x}{40}\right)$

On comparing with $y = x\tan\theta\left(1 - \frac{x}{R}\right)$

Horizontal range R = 40 m

For maximum height, $y = h_{\max}$ if $x = \frac{R}{2}$



$\therefore\ y = x\left(1 - \frac{x}{40}\right)$

Putting $x = \frac{R}{2} = 20\,\text{m}$

$y = h_{\max} = 20\left(1 - \frac{20}{40}\right) = 10\,\text{m}$

(NCERT Ed. 2025-26, 11th, Page No. 39)

38. (1)

Net force on the system is constant, so momentum is conserved.

$p_i = p_f$

$0 = m_1u_1 + m_2u_2$

$\frac{u_1}{u_2} = -\frac{m_2}{m_1} = -\left[\frac{1}{2}\right]$ as $\frac{m_1}{m_2} = \frac{2}{1}$

move in opposite direction with speed ratio 1:2.

(NCERT Ed. 2025-26, 11th, Page No. 101)

39. (2)

Pitch $= \frac{0.2}{2} = 0.1\,\text{cm} = 1\ \text{mm}$

$LC = \frac{\text{Pitch}}{\text{No. of CSD}} = \frac{1}{50} = 0.02\ \text{mm}$

(NCERT Ed. 2025-26, 11th, Page No. 11)

[Contd...

40. (3)

Slope of a-x graph = – tanθ

we know, $a = -\omega^2 x \Rightarrow \tan\theta = \omega^2$

$$\tan\theta = \left(\frac{2\pi}{T}\right)^2$$

$$\tan\theta = \left(\frac{2\pi}{2\pi}\right)^2 = 1$$

$\therefore\ \theta = 45°$

(NCERT Ed. 2025-26, 11th, Page No. 272)

41. (4)

Since speed is constant, the upward driving force must balance all downward forces (gravity + friction).

Total Force

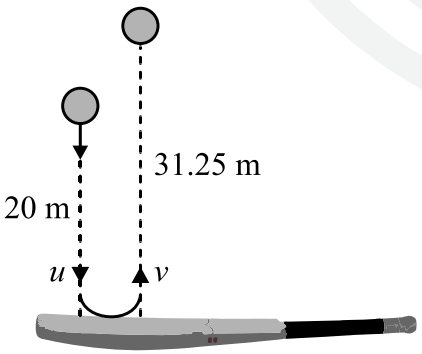
$$F = Mg + f = (1200 \times 10) + 3000 = 15000 \text{ N}.$$

Power

$$P = F \times v = 15000 \times 2.5 = 37500 \text{ W} = 37.5 \text{ kW}.$$

(NCERT Ed. 2025-26, 11th, Page No. 83)

42. (2)



$$|\overrightarrow{\text{Impulse}}| = m(\vec{v} - \vec{u})$$

$$= 0.2\left(\sqrt{2g(31.25)} + \sqrt{2g(20)}\right)$$

$$= 0.2\left(\sqrt{625} + \sqrt{400}\right)$$

= 0.2 × 45 = 9 kg/ms

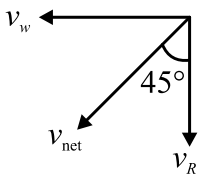
(NCERT Ed. 2025-26, 11th, Page No. 55)

43. (1)

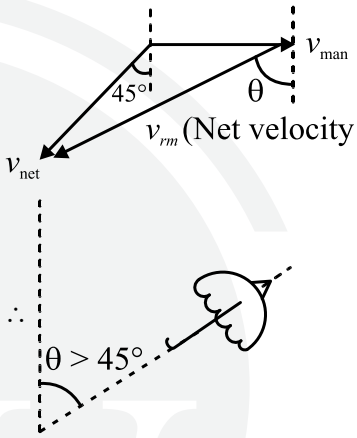
In case (B), the lens is made up of two materials vertically which leads to converging of incident rays at two different positions, hence forming two images.

(NCERT Ed. 2025-26, 12th, Page No. 238)

44. (3)



(wrt ground)



(NCERT Ed. 2025-26, 11th, Page No. 31)

45. (2)

Identify the positions of the individual centers of mass:

Let the center of the disc be at the origin $(0,0)$. Mass $m_1 = 2M$.

Since the ring has radius R and passes through the center of the disc, its own center must be at a distance from the origin. Let its center be at $(R,0)$

Mass $m_2 = M$.

Apply the Center of Mass formula:

$$X_{cm} = \frac{m_1 x_1 + m_2 x_2}{m_1 + m_2}$$

$$X_{cm} = \frac{(2M \times 0) + (M \times R)}{2M + M}$$

$$X_{cm} = \frac{MR}{3M} = \frac{R}{3}$$

(NCERT Ed. 2025-26, 11th, Page No. 96)

[Contd...

46. (1)

$\ln k = Ae^{-E_a/RT}$, A has the same units as k (true), and at a fixed T a smaller E_a gives a larger k (faster reaction). Statement I is too restrictive; IV and V are not true in the Arrhenius treatment.

(NCERT Ed. 2025-26, 12th, Page No. 83)

47. (3)

In Fischer projections with lowest priority (H) on the vertical (away):

Left structure: priorities OH > COOH > CH_3 > H; 1→2→3 goes clockwise → R.

Right structure: priorities NH_2 > COOH > C_2H_5 > H; 1→2→3 goes counterclockwise → S.

(NCERT Ed. 2025-26, 12th, Page No. 178)

48. (4)

In $[PtCl_3(C_2H_4)]^-$ the ligands bound to Pt are Cl^- (three of them, same type) and ethene (η^2-alkene). The outside water is just hydration, not coordinated.

(NCERT Ed. 2025-26, 12th, Page No. 124)

49. (3)

Low-spin d^5 octahedral → $t_{2g}^5 e_g^0$ with one unpaired e^-;

spin-only moment $\mu_{so} = \sqrt{n(n+2)}$

$= \sqrt{1 \cdot 3} \approx 1.73 BM$

(NCERT Ed. 2025-26, 12th, Page No. 134)

50. (1)

Ethyl phenyl ether is made by Williamson synthesis: (S_N2 on the alkyl halide).

But bromobenzene + NaOEt doesn't work—aryl halides don't undergo S_N2 to give ether.

(NCERT Ed. 2025-26, 12th, Page No. 216)

51. (2)

Serine has –OH (A–II); Methionine is an essential amino acid (C–I); Glycine is optically inactive (achiral) (D–III). Thus B maps to the remaining option glutamic acid which is a acidic amino acid (B–IV).

(NCERT Ed. 2025-26, 12th, Page No. 292)

52. (1)

First-order decay: $[A]_t = [A]_0 (1/2)^{t/t_{1/2}}$.

So after 900 s:

$[A] = A_0 (1/2)^{900/300} = A_0/8$,

$[B] = B_0 (1/2)^{900/180} = B_0/32$.

With $A_0 = B_0$, ratio $= (A_0/8)/(B_0/32) = 4$.

(NCERT Ed. 2025-26, 12th, Page No. 76)

53. (2)

Many coordination compounds are neutral (e.g., $[PtCl_2(NH_3)_2]$) and don't produce ions on dissolving. However, a complex ion (inner coordination sphere) generally remains intact in solution and doesn't dissociate into free metal and ligands.

(NCERT Ed. 2025-26, 12th, Page No. 123)

54. (1)

Zero-order kinetics arise from complex/surface-saturated or photochemical mechanisms; an elementary single-step reaction cannot be zero order (rate of an elementary step depends on at least one concentration).

(NCERT Ed. 2025-26, 12th, Page No. 80)

55. (3)

CO is a strong-field ligand because of π-acceptor (back-bonding) ability arising from its π/π* system, which stabilizes metal–ligand bonding; Cl^- lacks this.

(NCERT Ed. 2025-26, 12th, Page No. 138)

56. (2)

Number of angular nodes = l

s: $l = 0 \rightarrow 0$; p: $l = 1 \rightarrow 1$;

d: $l = 2 \rightarrow 2$; f: $l = 3 \rightarrow 3$

So $2s \rightarrow 0$ (IV), $5f \rightarrow 3$ (I),

$4d \rightarrow 2$ (III), $3p \rightarrow 1$ (II)

(NCERT Ed. 2025-26, 11th, Page No. 59)

57. (4)

$K_p / K_c = (RT)^{\Delta n_g}$

(A) $\Delta n_g = 2 \rightarrow (RT)^2$ (IV)

(B) $\Delta n_g = -1 \rightarrow 1/RT$ (III)

(C) $\Delta n_g = +1 \rightarrow RT$ (I)

(D) $\Delta n_g = -2 \rightarrow 1/(RT)^2$ (II)

(NCERT Ed. 2025-26, 11th, Page No. 178)

58. (3)

H_2 has configuration $(\sigma_{1s})^2$. Ionization removes one electron from the σ bonding MO → $(\sigma_{1s})^1$ in H_2^+, leaving one unpaired electron ⇒ paramagnetic.

(NCERT Ed. 2025-26, 11th, Page No. 129)

59. (2)

Let total moles = 1 → $n_s = 0.009$,

$n_w = 0.991$

Molality $m = \dfrac{n_s}{(\text{kg of water})}$

$= \dfrac{0.009}{0.991 \times 18/1000} \approx 0.50 m$

(NCERT Ed. 2025-26, 12th, Page No. 5)

[Contd...

60. (3)
Propan-1-ol forms intermolecular H-bonding (–OH), giving much stronger attractions and thus a higher b.p. than non-polar propane; R correctly explains A.
(NCERT Ed. 2025-26, 12th, Page No. 200)

61. (2)
1 L solution has mass $1.96\times10^{3}=1960$ g; at 29% w/w, (H_2SO_4) = 0.29 × 1960 = 568.4 g → moles $=568.4/98=5.8$. Hence $M=5.8\ \text{mol L}^{-1}$.
(NCERT Ed. 2025-26, 12th, Page No. 4)

62. (4)
Z = 108 : Unniloctium (Uno)
(NCERT Ed. 2025-26, 11th, Page No. 80)

63. (1)
Moles in 200 mg CO_2
$=0.200/44=4.545\times10^{-3}$
Molecules $=4.545\times10^{-3}\times N_A\approx2.73\times10^{21}$.
Left after removing $1.204\times10^{21}=$
$2.73-1.204\approx1.53\times10^{21}$.
(NCERT Ed. 2025-26, 11th, Page No. 18)

64. (2)
For non-relativistic particles
$\lambda=\dfrac{h}{\sqrt{2mK.E.}}\Rightarrow\lambda\propto1/\sqrt{K.E.}$.
If K.E. becomes 9 time of original, λ becomes $1/\sqrt{9}=1/3$ of original.
(NCERT Ed. 2025-26, 11th, Page No. 50)

65. (3)
$(I.E.)_1$ drops sharply from B (≈801 kJ/mol) to Al (≈578 kJ/mol), but from Al to Ga there is essentially no drop (Ga ≈ 579 kJ/mol) due to poor shielding by the filled $3d^{10}$ subshell in Ga, which increases effective nuclear charge.
(NCERT Ed. 2022-23, 11th, Page No. 317)

66. (1)
Down a group, atomic radius increases while electronegativity decreases (due to added shells and poorer effective nuclear attraction).
(NCERT Ed. 2025-26, 11th, Page No. 91)

67. (1)
These have a lone pair on the central atom (AX_3E) → trigonal pyramidal and non-planar (VSEPR). Carbonate (CO_3^{2-}) and borate (BO_3^{3-}) are trigonal planar.
(NCERT Ed. 2025-26, 11th, Page No. 115)

68. (1)
In acidic medium, manganate(VI) (MnO_4^{2-}) disproportionates to permanganate(VII) and manganese dioxide:
$3MnO_4^{2-}+4H^+\rightarrow2MnO_4^-+MnO_2+2H_2O$ giving Mn in +7 and +4 oxidation states.
(NCERT Ed. 2025-26, 12th, Page No. 106)

69. (1)
$$SO_2+Ca(OH)_2\rightarrow\underset{\text{milky}}{CaSO_3}+H_2O$$

70. (2)
For $AB_2(s)\rightleftharpoons A^{2+}+2B^-$, $K_{sp}=[A^{2+}][B^-]^2$.
So $\left[B^-\right]^2=\dfrac{6.91\times10^{-12}}{1.2\times10^{-4}}=5.76\times10^{-8}$
$\Rightarrow[B^-]\approx2.4\times10^{-4}\text{ M}$
(NCERT Ed. 2025-26, 11th, Page No. 204)

71. (1)
HCl adds to 3-chloro-1-butene give 2,3-dichlorobutane, which has two chiral centers → three stereoisomers: a meso form (inactive) and an enantiomeric pair (2 optically active).
(NCERT Ed. 2025-26, 11th, Page No. 311)

72. (1)
Add Br_2 → vic-dibromide; alc KOH then $NaNH_2$ eliminate 2 HBr → alkyne; $Na/NH_3(l)$ (dissolving-metal reduction) reduces alkyne to the trans-alkene.
(NCERT Ed. 2025-26, 11th, Page No. 315)

73. (2)
$$C_2H_5-C\equiv CH\xrightarrow[HgSO_4/H_2SO_4]{H_2O}C_2H_5-\underset{\underset{OH}{|}}{C}=CH_2\rightleftharpoons C_2H_5COCH_3$$
(NCERT Ed. 2025-26 11th, Page No. 317)

74. (1)
For a non-volatile solute, relative lowering $=\dfrac{p^0-p}{p^0}=\chi_{solute}=\dfrac{n_2}{n_1+n_2}$.
With $n_1=5.0$ mol and
$\chi_{solute}=1/4:\ \dfrac{n_2}{5+n_2}=1/4\Rightarrow n_2=5/3$ mol.
(NCERT Ed. 2025-26, 12th, Page No. 16)

75. (4)
2,2-Dimethylpropanal lacks any α-H, so it undergoes Cannizzaro in conc. NaOH (true). Cannizzaro is indeed a disproportionation (true) but that fact doesn't explain why this aldehyde reacts—absence of α-H does.
(NCERT Ed. 2025-26, 12th, Page No. 246)

[Contd...

76. (1)

Relative (and %) lowering $= \chi_{solute} = \frac{n_2}{n_1 + n_2}$.

$n_2 = 60/120 = 0.5$ mol; $n_1 = 180/18 = 10$ mol $\rightarrow$

$\chi_2 = \frac{0.5}{10.5} = 0.0476 \Rightarrow 4.76\%$.

(NCERT Ed. 2025-26, 12th, Page No. 16)

77. (1)

$\Delta G^\circ = -nFE^\circ = -RT\ln K \Rightarrow \log K = \frac{nFE^\circ}{2.303RT}$

At 298 K with $n = 2$ and $E^\circ = 0.177$V:

$\log K \approx \frac{2 \cdot 96485 \cdot 0.177}{2.303 \cdot 8.314 \cdot 298} \approx 6 \Rightarrow K \approx 10^6$.

(NCERT Ed. 2025-26, 12th, Page No. 39)

78. (1)

Xe has 8 valence e^-; in XeF_6 it uses 6 for Xe–F bonds, leaving 1 lone pair $\rightarrow$ steric number 7 $\Rightarrow$ sp^3d^3 hybridization (distorted octahedral).

(NCERT Ed. 2022-23, 12th, Page No. 210)

79. (3)

Alkene stability increases with degree of substitution due to hyperconjugation (more adjacent C–H bonds donate).

Hence tetra-substituted > tri-substituted > mono-substituted:

2,3-dimethylbut-2-ene > 2-methylbut-2-ene > but-1-ene. R correctly explains A.

(NCERT Ed. 2025-26, 11th, Page No. 277)

80. (1)

Eu^{2+}: $[Xe]4f^7$ $\rightarrow$ 7 unpaired (half-filled f).

Ce^{3+}: $[Xe]4f^1$ $\rightarrow$ 1 unpaired.

(NCERT Ed. 2025-26, 12th, Page No. 108)

81. (4)

(I) has one Me and one Et substituent on N $\rightarrow$ ethylmethylamine (B).

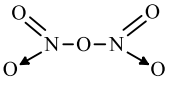
(II) $CH_3CH_2CH_2NH_2$ $\rightarrow$ n-propylamine (D).

(III) $(CH_3)_2CHNH_2$ $\rightarrow$ isopropylamine (C).

(IV) N attached to Et and two Me groups $\rightarrow$ N,N-dimethylethylamine (A).

(NCERT Ed. 2025-26, 12th, Page No. 261)

82. (2)

- In solid state PCl_5 exist as an ionic solid, $[PCl_4]^+[PCl_6]^-$.
- N_2O_5: 

83. (4)

First step gives chlorobenzene. Halogens are deactivating but o/p-directing; nitration therefore occurs at ortho and para positions $\rightarrow$ mixture of o- and p-nitrochlorobenzene.

(NCERT Ed. 2025-26, 12th, Page No. 185)

84. (3)

Kjeldahl method estimates N that forms ammonium sulfate on digestion. Nitro N (as in nitrobenzene) does not convert to NH_4^+ under these conditions; hence its nitrogen cannot be determined by this method—R correctly explains A.

(NCERT Ed. 2025-26, 11th, Page No. 288)

85. (3)

(A) (C_6H_5COOH $\rightarrow$ [$SOCl_2$] C_6H_5COCl $\rightarrow$ [H_2/Pd-$BaSO_4$] C_6H_5CHO)(Rosenmund).

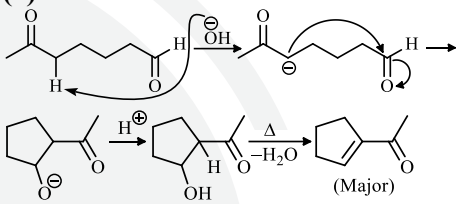
(B) ($C_6H_5CH_2OH$) with (CrO_3/H_2SO_4 $\rightarrow$ benzoic acid.

(C) $NaBH_4$ does not reduce esters; the sequence won't give benzaldehyde.

(D) ($C_6H_5CH_3$ $\rightarrow$[$CrO_3/(CH_3CO)_2O$] [H_3O + Δ] C_6H_5CHO) (Etard via chromyl acetate).

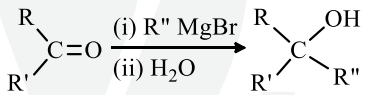
(NCERT Ed. 2025-26, 12th, Page No. 233)

86. (1)



(NCERT Ed. 2025-26, 12th, Page No. 245)

87. (4)



(NCERT Ed. 2025-26 12th, Page No. 201)

88. (1)

These are water-soluble vitamins and are not stored extensively; A, D, E, K are fat-soluble and stored in the body.

(NCERT Ed. 2025-26, 12th, Page No. 296)

89. (1)

$CHCl_3$/KOH converts a 1° aryl amine to the isocyanide (carbylamine). Catalytic hydrogenation (Pd/C, H_2) reduces –NC to $–NHCH_3$ and also reduces the aryl ketone to the benzylic alcohol.

(NCERT Ed. 2025-26, 12th, Page No. 272)

90. (2)

A. Ester hydrolysis $\xrightarrow{H_2SO_4/H_2O}$ ethanol (II).

B. Ester $\xrightarrow{DIBAL-H/H_2O}$ aldehyde (III).

C. Nitrile $\xrightarrow{SnCl_2, HCl/H_2O}$ (Stephen) $\rightarrow$ aldehyde (IV).

D. Nitrile $\xrightarrow{CH_3MgBr/H_3O^+}$ ketone (I).

(NCERT Ed. 2025-26, 12th, Page No. 242)

[Contd...

91. (2)

List-I	List-II
E	Represent the genus of source bacteria
co	Represent the species of source bacteria
R	Name of strain
I	Indicate the order in which the enzymes were isolated from that strain of bacteria

(NCERT Ed. 2025-26, 12th, Page No. 165)

92. (3)

RNA polymerase III is responsible for transcription of tRNA, 5srRNA and snRNAs in eukaryotes. tRNA is essential for reading the genetic code and bringing the correct amino acids to the ribosome during translation (synthesis of polypetides). If RNA polymerase III is defective, the cell cannot produce tRNAs. Without tRNAs, the "translation machinery" cannot function, and the synthesis of polypeptides (proteins) in the cytoplasm will stop.

(NCERT Ed. 2025-26, 12th, Page No. 95, 99)

93. (2)

The generative cell divides mitotically to give rise to the two male gametes in the pollen grain. There is a characteristic distribution of the cells within the embryo sac. Three cells are grouped together at the micropylar end and constitute the egg apparatus. The egg apparatus, in turn, consists of two synergids and one egg cell. The synergids have special cellular thickenings at the micropylar tip called filiform apparatus, which play an important role in guiding the pollen tubes into the synergid. Three cells are at the chalazal end and are called the antipodals. The large central cell has two polar nuclei. Thus, a typical angiosperm embryo sac, at maturity, though 8-nucleate is 7-celled.

(NCERT Ed. 2025-26, 12th, Page No. 11)

94. (2)

In basidiomycetes, the mycelium is branched and septate. The asexual spores are generally not found, but vegetative reproduction by fragmentation is common.

The sex organs are absent, but plasmogamy is brought about by fusion of two vegetative or somatic cells of different strains or genotypes.

The resultant structure is dikaryotic which ultimately gives rise to basidium. Karyogamy and meiosis take place in the basidium producing four basidiospores. The basidiospores are exogenously produced on the basidium. Vegetative reproduction by fragmentation is common.

(NCERT Ed. 2025-26, 11th, Page No. 18)

95. (2)

The hypothalamus is the basal part of the forebrain and it regulates a wide spectrum of body functions. It contains several groups of neurosecretory cells called nuclei which produce hormones. These hormones regulate the synthesis and secretion of pituitary hormones. Somatostatin from the hypothalamus inhibits the release of growth hormone from the pituitary.

(NCERT Ed. 2025-26, 11th, Page No. 240)

96. (2)

Bt toxin is produced by a bacterium called *Bacillus thuringiensis* (*Bt* for short). *Bt* toxin gene has been cloned from the bacteria and been expressed in plants to provide resistance to insects without the need for insecticides; in effect created a bio-pesticide.

(NCERT Ed. 2025-26, 12th, Page No. 179)

97. (3)

Mesosomes help in cell wall formation, DNA replication and distribution to daughter cells. They also help in respiration, secretion processes, to increase the surface area of the plasma membrane and enzymatic content.

(NCERT Ed. 2025-26, 11th, Page No. 91)

98. (2)

The decomposition is quicker if detritus is rich in nitrogen/sugars and if the environment is warm and moist. Setup I + Detritus B provides the ideal combination of "quality" and "climate." Option 1 is incorrect because lignin significantly slows decomposition even in good climates. Option 3 and 4 is incorrect because anaerobiosis and low temperature inhibit decomposition (leading to buildup, not necessarily faster processing into humus).

(NCERT Ed. 2025-26, 12th, Page No. 208)

99. (2)

Bacillus thuringiensis (often written as *Bt*). These are available in sachets as dried spores which are mixed with water and sprayed onto vulnerable plants such as brassicas and fruit trees, where these are eaten by the insect larvae. In the gut of the larvae, the toxin is released and the larvae get killed. The bacterial disease will kill the caterpillars, but leave other insects unharmed.

(NCERT Ed. 2025-26, 12th, Page No. 157)

100. (2)

The features of wind-pollinated flowers are light/non-sticky pollen, well-exposed stamens, and feathery stigma. Both wind and water pollinated flowers are not very colourful and do not produce nectar. Therefore, large colourful flowers and production of nectar are not associated with them.

(NCERT Ed. 2025-26, 12th, Page No. 13)

101. (2)

List-I	List-II
Origin of universe	13.8 bya
Origin of earth	4.5 bya
Origin of life on earth	4 bya
First cellular form of life	2000 mya

(NCERT Ed. 2025-26, 12th, Page No. 111)

102. (1)

The transition state is an unstable and intermediate structural state.

During this, substrate bonds are broken and new bonds are established that transform the substrate molecules into product.

This state is transient and highly unstable.

(NCERT Ed. 2025-26, 11th, Page No. 115)

103. (4)

DNA fragments are negatively charged molecules they can be separated by forcing them to move towards the anode under an electric field through a medium/matrix. Nowadays, the most commonly used matrix is agarose which is a natural polymer extracted from sea weeds. The DNA fragments separate (resolve) according to their size through the sieving effect provided by the agarose gel. Hence, the smaller the fragment size, the farther it moves.

(NCERT Ed. 2025-26, 12th, Page No. 168)

104. (2)

Traits expressed in both homozygous and heterozygous conditions are dominant ones.

- Axial flower position in pea plant – dominant
- Violet flower colour in pea plant– dominant
- Round seed shape in pea plant – dominant
- Pink colour flower in snapdragon = incomplete dominance
- Green seed colour in pea plant = recessive

(NCERT Ed. 2025-26, 12th, Page No. 54)

105. (2)

In a normal ECG, the QRS complex represents the depolarisation of the ventricles, which initiates the ventricular contraction.

(NCERT Ed. 2025-26, 11th, Page No. 201)

106. (2)

List-I	List-II
Amniocentesis	Detection of genetic disorder in unborn foetus
Lactational amenorrhea	Absence of menstruation due to intense lactation
MTP	Induced abortion
ART	Assist those couple which are childless

(NCERT Ed. 2025-26, 12th, Page No. 46)

107. (1)

List-I	List-II
Fighting fish	*Betta*
Angel fish	*Pterophyllum*
Flying fish	*Exocoetus*
Sea horse	*Hippocampus*

(NCERT Ed. 2025-26, 11th, Page No. 48)

108. (1)

List-I (Hormone)	List-II (Function)
Parathyroid hormone (PTH)	Increase blood calcium levels
Aldosterone	Regulates water and electrolyte balance
Adrenal Androgens	Growth of axial, pubic, and facial hair during puberty
Cortisol	Produces anti-inflammatory reactions and suppresses immune response

(NCERT Ed. 2025-26, 11th, Page No. 245)

109. (2)

List-I	List-II
Squamous epithelium	Formation of diffusion boundary
Cuboidal epithelium	Secretion and absorption
Compound epithelium	Protection against chemical and mechanical stresses
Ciliated epithelium	Movement of particle in specific direction

(NCERT Ed. 2020-21, 11th, Page No. 101)

[Contd...

110. (2)

Non-glandular skin lacks glands that produce mucus or other secretions. Reptiles, such as *Chameleon* and turtles, have dry, non-glandular skin covered by scales or scutes.

(NCERT Ed. 2025-26, 11th, Page No. 49)

111. (4)

The medulla contains centres which control respiration, cardiovascular reflexes and gastric secretions. The hypothalamus contains a number of centres which control body temperature, urge for eating and drinking.

(NCERT Ed. 2025-26, 11th, Page No. 235)

112. (4)

List-I	List-II
Cervical	7 in number
Thoracic	12 in number
Coccygeal	1 in number after fusion
Lumber	5 in number

(NCERT Ed. 2025-26, 11th, Page No. 225)

113. (3)

List-I	List-II
Heroin	Depressant
Datura	Hallucinogen
Cocaine	Coke
Morphine	Pain killer

(NCERT Ed. 2025-26, 12th, Page No. 143)

114. (4)

Coronary Artery Disease, often referred to as atherosclerosis, affects the vessels that supply blood to the heart muscle. It is caused by deposits of calcium, fat, cholesterol and fibrous tissues, which makes the lumen of arteries narrower.

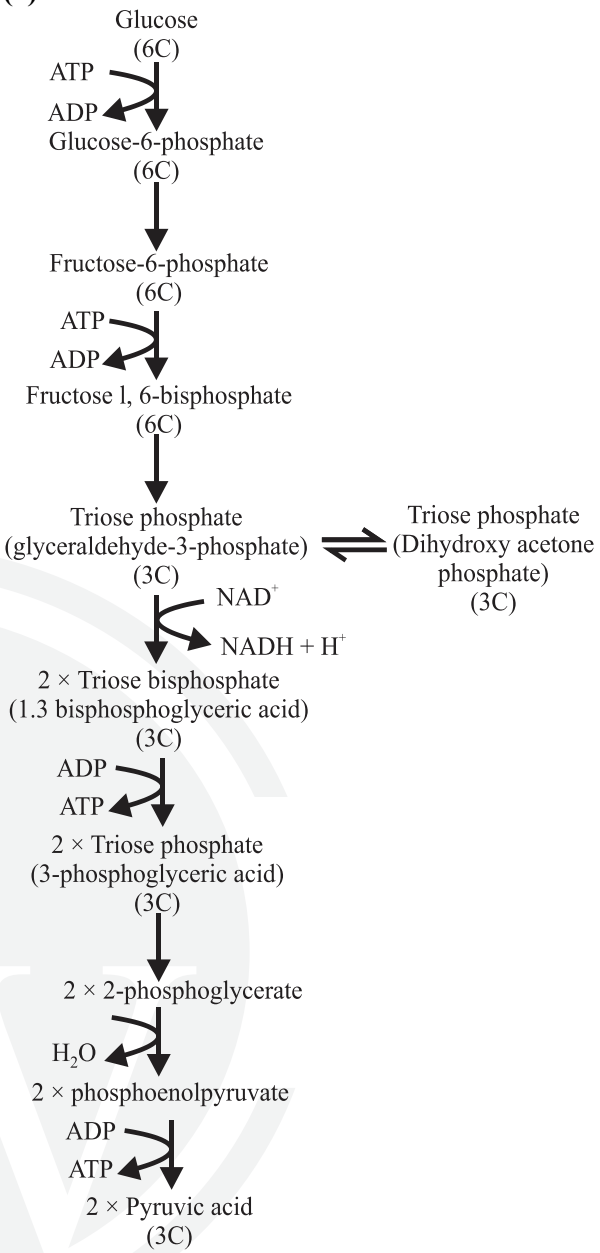
(NCERT Ed. 2025-26, 11th, Page No. 203)

115. (3)

In honey bees an offspring formed from the union of a sperm and an egg develops as a female (queen or worker), and an unfertilized egg develops as a male (drone) by means of parthenogenesis. This means that the males have half the number of chromosomes than that of a female. The females are diploid having 32 chromosomes and males are haploid, i.e., having 16 chromosomes.

(NCERT Ed. 2025-26, 12th, Page No. 71)

116. (1)



(NCERT Ed. 2025-26, 11th, Page No. 156, 157)

117. (4)

Genotypes:

Female → X^CX^C

Male →X^CY

Punnett square:

	X^C (male)	Y (male)
X^C	$X^C X^C$	X^CY
X^C	$X^C X^C$	X^CY

Daughters: X^CX^C → 100% colour-blind

Sons X^CY → 100% colour-blind

(NCERT Ed. 2025-26, 12th, Page No. 73)

118. (4)

The Indian Parliament has recently cleared the second amendment of the Indian Patents Bill, that takes such issues into consideration, including patent terms, emergency provisions and research and development initiative.

(NCERT Ed. 2025-26, 12th, Page No. 185)

[Contd...

119. (1)

- Monera exhibits the most diverse nutrition (Photosynthetic, Chemosynthetic, Heterotrophic).
- Protista includes both autotrophic (e.g., Diatoms) and heterotrophic (e.g., Protozoans) members.
- Fungi are exclusively heterotrophic with absorptive nutrition. Plantae are exclusively autotrophic (photosynthetic) except a few. Members of kingdom.
- Animalia are exclusively heterotrophic with ingestive nutrition.

(NCERT Ed. 2025-26, 11th, Page No. 18)

120. (2)

Monocot stem: Vascular bundles are conjoint and closed. The phloem parenchyma is absent, and water-containing cavities are present within the vascular bundles.

(NCERT Ed. 2025-26, 11th, Page No. 76)

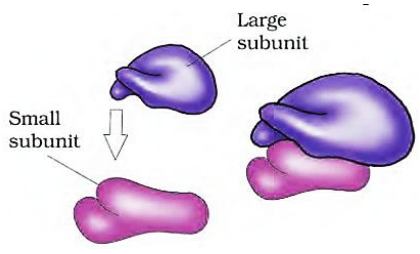
121. (4)

Due to cancer of the thyroid gland or due to development of nodules of the thyroid glands, the rate of synthesis and secretion of the thyroid hormones is increased to abnormally high levels leading to a condition called hyperthyroidism which adversely affects the body physiology.

(NCERT Ed. 2025-26, 11th, Page No. 243)

122. (2)

Ribosomes are non-membrane bound organelles found in all cells – both eukaryotic as well as prokaryotic. Ribosomes are the granular structures first observed under the electron microscope as dense particles by George Palade (1953). They are composed of ribonucleic acid (RNA) and proteins and are not surrounded by any membrane. The eukaryotic ribosomes are 80S while the prokaryotic ribosomes are 70S. Each ribosome has two subunits, larger and smaller subunits. The two subunits of 80S ribosomes are 60S and 40S while that of 70S ribosomes are 50S and 30S.



Ribosome

(NCERT Ed. 2025-26, 11th, Page No. 98)

123. (4)

The primary transcript (mRNA) is synthesised complementary to the template (antisense) strand of DNA, and similar to coding strand with uracil (U) replacing thymine (T).

To find the primary transcript, replace T with U 5' GCAUGCAUGCAUU3'

(NCERT Ed. 2025-26, 12th, Page No. 92)

124. (4)

Cerebellum not pons has a very convoluted surface in order to provide the additional space for many more neurons. Pons consists of fibre tracts that interconnect different regions of the brain. Three major regions make up the brain stem; mid brain, pons and medulla oblongata.

(NCERT Ed. 2025-26, 11th, Page No. 236)

125. (4)

The alveoli and their ducts form the respiratory or exchange part of the respiratory system in humans. The exchange part is the site of actual diffusion of O_2 and CO_2 between blood and atmospheric air.

(NCERT Ed. 2025-26, 11th, Page No. 185)

126. (2)

Evolutionary Biology is the study of the history of life forms on the earth.

The Big Bang theory attempts to explain to us the origin of universe.

(NCERT Ed. 2025-26, 12th, Page No. 110)

127. (1)

Chara	Green alga	Chloroplasts contain pyrenoids, chlorophyll *a* and *b*
Anabaena	Cyanobacterium	Chloroplasts are not present, can fix atmospheric nitrogen in specialised cells
Polysiphonia	Red alga	Oogamous sexual reproduction, chloroplasts contain *a* and *d*
Dictyota	Brown alga	Stored food is laminarin or mannitol, chloroplasts contain *a* and *c*

(NCERT Ed. 2025-26, 11th, Page No. 27)

[Contd...

128. (3)

Malvaceae is commonly called the cotton family. The fruit in members of family Graminae is generally a caryopsis.

(NCERT Ed. 2025-26, 11th, Page No. 64)

129. (4)

Cytokinins help overcome the apical dominance.

(NCERT Ed. 2025-26, 11th, Page No. 177)

130. (2)

Both C_3 and C_4 plants use the Calvin cycle to synthesize sugar (glucose). In C plants, the initial CO_2 fixation occurs in the mesophyll cells, but the final fixation and sugar synthesis via the Calvin cycle take place in the bundle sheath cells. In C_3 plants, the primary CO_2 acceptor is RuBP (a 5-carbon compound). In C_4 plants, the primary acceptor is PEP (a 3-carbon compound). Kranz anatomy is a specialized leaf structure found only in C_4 plants, where bundle sheath cells are arranged in a wreath-like manner. Generally, C_4 plants are more efficient at photosynthesis in high light and temperature conditions and typically produce more biomass compared to C_3 plants, which lose energy through a process called photorespiration.

(NCERT Ed. 2025-26, 11th, Page No. 147)

131. (2)

Sperm in frog forms in testes from there it moves to vasa efferentia. From vasa efferentia they enter the kidneys on their side and open into Bidder's canal. Finally it communicates with the urinogenital duct that comes out of the kidneys and opens into the cloaca. The cloaca is a small, median chamber that is used to pass faecal matter, urine and sperms to the exterior through cloacal aperture.

(NCERT Ed. 2025-26, 11th, Page No. 84)

132. (3)

Proximal Convoluted Tubule (PCT) is lined by a simple cuboidal brush border epithelium which increases the surface area for reabsorption. Nearly all of the essential nutrients including glucose, and 70-80 per cent of electrolytes and water are reabsorbed by this segment. Glucose is reabsorbed actively by this segment.

(NCERT Ed. 2025-26, 11th, Page No. 209)

133. (3)

Proper information about reproductive organs, adolescence and related changes, safe and hygienic sexual practices, sexually transmitted diseases (STD), AIDS, etc., would help people, especially those in the adolescent age group to lead a reproductively healthy life.

(NCERT Ed. 2025-26, 12th, Page No. 42)

134. (4)

Cryopreservation is an example of *ex-situ conservation*.

(NCERT Ed. 2025-26, 12th, Page No. 225)

135. (1)

Ecology is one such thread which gives us a holistic perspective to biology. A particular aspect of this is the study of anthropogenic environmental degradation and the socio-political issues it has raised.

(NCERT Ed. 2025-26, 12th, Page No. 189)

136. (2)

Symptoms of ascariasis include internal bleeding, muscular pain, fever, anemia and blockage of the intestinal passage.

(NCERT Ed. 2025-26, 12th, Page No. 133)

137. (2)

The gametes in brown algae are pyriform (pear-shaped) and bear two laterally attached flagella (phaeophyceae). Sexual reproduction is oogamous and accompanied by complex post-fertilisation developments in red algae (rhodophyceae).

(NCERT Ed. 2025-26, 11th, Page No. 27)

138. (3)

Given data:
Initial Population = 375
Births (B) = 72
Immigration (I) = 32
Deaths (D) = 40
Emigration (E) = 56
Total number of individuals added to population = 72 + 32 = 104
Total number of individuals removed from population = 56 + 40 = 96
Net change in population = 104-96 = increase in 8 individuals.

(NCERT Ed. 2025-26, 12th, Page No. 193)

[Contd...

139. (4)

The floral formula of Brassicaceae is

$\oplus\ K_{2+2}C_4A_{2+4}\underline{G}_{(2)}$

Tetradynamous Condition: This specific arrangement of six stamens is a unique and diagnostic feature of the Brassicaceae family (formerly known as Cruciferae). The stamens are arranged in two whorls:

An outer whorl with two shorter lateral stamens.

An inner whorl with four longer medial stamens.

Floral Formula Representation: In the floral formula, this condition is represented as A_{2+4}, where 'A' stands for androecium (male reproductive part/stamens), the subscript indicates the number and arrangement of stamens (2 in one whorl, 4 in another).

Solanaceae (e.g., tomato, potato) typically has five epipetalous stamens (fused to the petals) of equal length.

Malvaceae (e.g., cotton, *Hibiscus*) typically has numerous stamens united into a single tube (monoadelphous condition).

Graminae (Poaceae, grass family) have a very different flower structure, usually with three versatile stamens and a highly reduced perianth.

(NCERT Ed. 2025-26, 11th, Page No. 64, 68)

140. (4)

*Eco*RI recognizes and cuts the DNA sequence 5'-GAATTC-3'. The complementary (antisense) strand of this sequence is the reverse complement of the recognition sequence.

The recognition sequence 5'-GAATTC-3' on one strand has its complementary sequence on the other strand, which is 3'-CTTAAG-5'.

(NCERT Ed. 2025-26, 12th, Page No. 166)

141. (3)

RrYy × rryy	1: 1 : 1: 1
RrYy × RrYy	9 : 3 : 3 : 1
Rr × Rr	3: 1
Rr × rr	1: 1

(NCERT Ed. 2025-26, 12th, Page No. 60, 64)

142. (4)

List-I	List-II
Parturition	Delivery of baby
Gestation	Period between fertilisation and birth
Ovulation	Release of egg
Implantation	Attachment of blastocyst to endometrium

(NCERT Ed. 2025-26, 12th, Page No. 27)

143. (4)

Although cell growth (in terms of cytoplasmic increase) is a continuous process, DNA synthesis occurs only during one specific stage in the cell cycle.

(NCERT Ed. 2025-26, 11th, Page No. 120)

144. (4)

The tangential as well as radial walls of the endodermal cells in roots have a deposition of water-impermeable, waxy material suberin in the form of casparian strips. In grasses, certain adaxial epidermal cells along the veins modify themselves into large, empty, colourless cells. These are called bulliform cells. The root hairs are unicellular elongations of the epidermal cells and help absorb water and minerals from the soil. Stomata regulate the process of transpiration and gaseous exchange.

(NCERT Ed. 2025-26, 11th, Page No. 74)

145. (2)

The plasma membrane of the red blood cells has sugar polymers that protrude from its surface and the kind of sugar is controlled by the gene. The gene (*I*) has three alleles I^A, I^B and *i*. The alleles I^A and I^B produce a slightly different form of the sugar while allele *i* does not produce any sugar. Since there are three different alleles, there are six different combinations of these three alleles that are possible, and therefore, a total of six different genotypes and four phenotypes of the human ABO blood types.

(NCERT Ed. 2025-26, 12th, Page No. 61)

146. (4)

Reduced levels of estrogen resulted in osteoporosis in old people. The number of ribs in humans is 12 pairs i.e. 24. Accumulation of uric acid crystals causes gout. Joints between Atlas and axis are an example of synovial joints. Pectoral and Pelvic girdle bones help in the articulation of the upper and the lower limbs respectively with the axial skeleton.

(NCERT Ed. 2025-26, 11th, Page No. 226)

147. (4)

Spermatids are haploid which contains 23 chromosomes. Secondary spermatocyte, secondary oocyte, ootid and ovum, all are haploid and contain 23 chromosomes.

(NCERT Ed. 2025-26, 12th, Page No. 31)

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48. (4)

The correct sequence in modifying an organism is:

(i) Identification of DNA with desirable genes.

(ii) Introduction of the identified DNA into the host.

(iii) Maintenance of introduced DNA in the host and transfer of the DNA to its progeny.

(NCERT Ed. 2025-26, 12th, Page No. 165)

[Contd...

149. (4)

Antonie Von Leeuwenhoek first saw and described a live cell. Robert Brown later discovered the nucleus. The invention of the microscope and its improvement leading to the electron microscope revealed all the structural details of the cell. In 1838, Matthias Schleiden, a German botanist, examined a large number of plants and observed that all plants are composed of different kinds of cells which form the tissues of the plant. At about the same time, Theodore Schwann (1839), a German Zoologist, studied different types of animal cells and reported that cells had a thin outer layer which is today known as the 'plasma membrane'. He also concluded, based on his studies on plant tissues, that the presence of cell wall is a unique character of the plant cells. Schleiden and Schwann together formulated the cell theory. This theory however, did not explain as to how new cells were formed. Rudolf Virchow (1855) first explained that cells divided and new cells are formed from pre-existing cells (*Omnis cellula-e cellula*). He modified the hypothesis of Schleiden and Schwann to give the cell theory a final shape.

(NCERT Ed. 2025-26, 11th, Page No. 88)

150. (4)

Cardiac arrest is the condition when the heart stops beating. In a heart attack, the heart muscle is suddenly damaged by an inadequate blood supply.

(NCERT Ed. 2025-26, 11th, Page No. 203)

151. (4)

The main criteria for classification used by Whittaker include cell structure, body organisation, mode of nutrition, reproduction and phylogenetic relationships. Let us look at this five kingdom classification to understand the issues and considerations that influenced the classification system. Earlier classification systems included bacteria, blue green algae, fungi, mosses, ferns, gymnosperms and the angiosperms under 'Plants'.

The character that unified this whole kingdom was that all the organisms included had a cell wall in their cells. This placed together groups which widely differed in other characteristics. It brought together the prokaryotic bacteria and the blue green algae (cyanobacteria) with other groups which were eukaryotic. It also grouped together the unicellular organisms and the multicellular ones, say, for example, *Chlamydomonas* and *Spirogyra* were placed together under algae. The classification did not differentiate between the heterotrophic group – fungi, and the autotrophic green plants, though they also showed a characteristic difference in their walls composition – the fungi had chitin in their walls while the green plants had a cellulosic cell wall. When such characteristics were considered, the fungi were placed in a separate kingdom – Kingdom Fungi.

(NCERT Ed. 2025-26, 11th, Page No. 12)

152. (1)

In the sliding filament theory of muscle contraction, the I-band (which contains only actin filaments) shortens as actin and myosin filaments slide past each other, while the A-band (which contains the entire length of myosin filaments) remains unchanged. The H-zone (the central region of the A-band, where only myosin is present) also shortens, but the filaments themselves do not change in length. The sliding mechanism causes the muscle to contract without altering the length of the actin or myosin filaments.

(NCERT Ed. 2025-26, 11th, Page No. 222)

153. (1)

The correct hierarchical sequence for *Homo sapiens* from species to kingdom is: *sapiens* → *Homo* → Hominidae → Primata → Mammalia → Chordata → Animalia.

(NCERT Ed. 2025-26, 11th, Page No. 8)

154. (1)

- Wheat belongs to class Monocotyledonae while mango belongs to class Dicotyledonae. Both belong to division angiospermae. *nigrum* and *melongena* both belong to genus *Solanum* and kingdom Plantae.
- *Petunia* and *Datura* are two different genera that belong to family Solanaceae. Man and housefly belong to kingdom Animalia but phylum Chordata and Arthropoda, respectively.

(NCERT Ed. 2025-26, 11th, Page No. 7)

155. (2)

In parietal placentation, the ovules develop on the inner wall of the ovary or on peripheral part. Ovary is one-chambered but it becomes two-chambered due to the formation of the false septum, e.g., mustard and *Argemone*. The label 'X' refers to the ovule that becomes seed after fertilisation. In tomato and lemon, axile placentation is seen. The placenta develops at the base of ovary in basal placentation.

(NCERT Ed. 2025-26, 11th, Page No. 65)

156. (4)

Leydig cells synthesise and secrete testicular hormones called androgens. FSH acts on the Sertoli cells and stimulates secretion of some factors which help in the process of spermiogenesis.

(NCERT Ed. 2025-26, 12th, Page No. 31)

157. (3)

The average composition of cells in humans is Water > Proteins > Nucleic acids > Carbohydrates > Lipids > Ions.

(NCERT Ed. 2025-26, 11th, Page No. 109)

158. (4)

The given diagram is of cotton destroyed by bollworms which can be controlled by the proteins encoded by the genes *cryIAc* and *cryIIAb*.

(NCERT Ed. 2025-26, 12th, Page No. 180)

159. (4)

The inner membrane potential changes from negative to positive, during depolarisation due to rapid influx of Na^+ ions. The electrical potential difference across the polarised plasma membrane is called the action potential. In a resting neuron, the axoplasm inside the axon contains high concentration of K^+.

(NCERT Ed. 2025-26, 11th, Page No. 233)

160. (1)

Adults of *Periplaneta americana* are about 34-53 mm long. In each segment of cockroach, exoskeleton has hardened plates called sclerites.

(NCERT Ed. 2020-21, 11th, Page No. 111)

161. (1)

The red algae reproduce asexually by non-motile spores and sexually by non-motile gametes. Union of gametes in brown algae may take place in water or within the oogonium (oogamous species). Most green algae have food stored in the form of starch. The cyanobacteria (blue-green algae) are classified as eubacteria (true bacteria).

(NCERT Ed. 2025-26, 11th, Page No. 27)

162. (4)

Today transgenic models exist for many human diseases such as cancer, cystic fibrosis, rheumatoid arthritis and Alzheimer's.

(NCERT Ed. 2025-26, 12th, Page No. 183)

163. (2)

IUDs are ideal contraceptives for the females who want to delay pregnancy and/or space children. It is one of most widely accepted methods of contraception in India.

(NCERT Ed. 2025-26, 12th, Page No. 45)

164. (1)

Colombia located near the equator has nearly 1,400 species of birds while New York at 41° N has 105 species and Greenland at 71° N only 56 species. India, with much of its land area in the tropical latitudes, has more than 1,200 species of birds.

(NCERT Ed. 2025-26, 12th, Page No. 219)

165. (3)

Lactose is the substrate for the enzyme beta-galactosidase and it regulates switching on and off of the *lac* operon. Hence, it is termed as inducer. The repressor of the operon is synthesised (all-the-time – constitutively) from the *i* gene. In the presence of an inducer, such as lactose or allolactose, the repressor is inactivated by interaction with the inducer. Regulation of *lac* operon by repressor is referred to as negative regulation.

(NCERT Ed. 2025-26, 12th, Page No. 100)

166. (3)

Lower invertebrates like sponges, coelenterates, flatworms, etc., exchange O_2 with CO_2 by simple diffusion over their entire body surface. Earthworms use their moist cuticle and insects have a network of tubes (tracheal tubes) to transport atmospheric air within the body.

We have one pair of external nostrils.

Workers in grinding and stone breaking industries suffer from lung fibrosis.

(NCERT Ed. 2025-26, 11th, Page No. 191)

167. (4)

List-I	List-II
Pigments	Carotenoids
Alkaloids	Morphine
Terpenoids	Monoterpenes
Essential oil	Lemon grass oil

(NCERT Ed. 2025-26, 11th, Page No. 108)

168. (2)

After entering one of the synergids, the pollen tube releases the two male gametes into the cytoplasm of the synergid. One of the male gametes moves towards the egg cell and fuses with its nucleus thus completing the syngamy. This results in the formation of a diploid cell, the zygote. The other male gamete moves towards the two polar nuclei located in the central cell and fuses with them to produce a triploid primary endosperm nucleus (PEN). In this case, the female plant is diploid (2n), so its polar nuclei are haploid (n). The male plant is tetraploid (4n), so its male gametes are diploid (2n). Therefore, Endosperm = fusion of one male gamete (2n) and two polar nuclei (n+n) = 2n + n + n = 4n.

(NCERT Ed. 2025-26, 12th, Page No. 18)

169. (2)

List-I	List-II
Bidder's canal	Present inside the kidney
Mesorchium	Connects testes to kidney
Cloaca	Common chamber for faeces and urine
Liver	Secretes bile

(NCERT Ed. 2025-26, 11th, Page No. 83)

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170. (2)

DNA is a hydrophilic molecule, it cannot pass through cell membranes. In a method known as micro-injection, recombinant DNA is directly injected into the nucleus of an animal cell.

(NCERT Ed. 2025-26, 12th, Page No. 170)

171. (3)

In amensalism one species is harmed whereas the other is unaffected. Both the species benefit in mutualism.

(NCERT Ed. 2025-26, 12th, Page No. 197)

172. (4)

The correct statement are:

A. Yoga has been practised to achieve physical and mental health.
B. Infectious diseases are very common and everyone of us suffers from these at sometime or other.
C. AIDS is an infectious disease.
D. Saliva in the mouth is an example of physiological barrier of innate immunity.
E. Healthy persons bring economic prosperity.

(NCERT Ed. 2025-26, 12th, Page No. 130)

173. (1)

In non-cyclic photophosphorylation (the Z-scheme), electrons travel from Photosystem I to Ferredoxin and then to FNR to reduce $NADP^+$ into NADPH. In cyclic photophosphorylation, electrons are cycled back from Ferredoxin to the Cytochrome B6f complex; therefore, FNR is not involved in the cyclic pathway. The stroma lamellae membranes possess PS I but lack PS II and the NADP reductase enzyme (FNR). FNR is not the primary electron acceptor from PS II, but is associated with the end of the electron transport chain after Photosystem I.

(NCERT Ed. 2025-26, 11th, Page No. 140)

174. (2)

There is a linear relationship between incident light and CO_2 fixation rates at low light intensities.

(NCERT Ed. 2025-26, 11th, Page No. 149)

175. (3)

Since both leaf A and B increased their surface area by the same total amount 5 cm^2, in the same time period, both have same absolute growth rate i.e. 5 cm^2 per day. Their relative growth rate is 0.1 and 0.05, respectively.

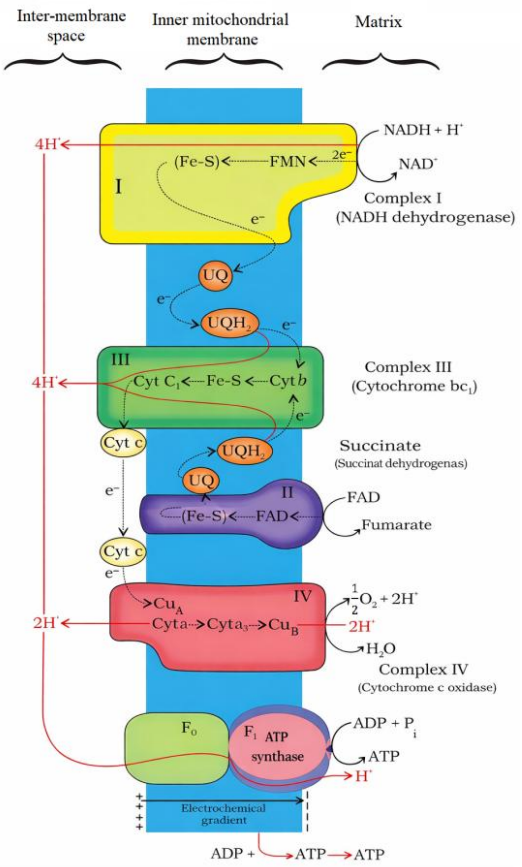
(NCERT Ed. 2025-26, 11th, Page No. 171)

176. (4)

Telophase I: The nuclear membrane and nucleolus reappear, cytokinesis follows and this is called as dyad of cells. Although in many cases the chromosomes do undergo some dispersion, they do not reach the extremely extended state of the interphase nucleus.

(NCERT Ed. 2025-26, 11th, Page No. 127)

177. (4)



Electron Transport System (ETS)

(NCERT Ed. 2025-26, 11th, Page No. 160)

178. (3)

Phloem transports food materials (photosynthates), usually from leaves to other parts of the plant.

(NCERT Ed. 2022-23, 11th, Page No. 87)

179. (1)

Oscillatoria	Biofertiliser
Nucleopolyhedrovirus	Bicontrol agent
Saccharomyces cerevisiae	Yeast used in making bread and beer
Monascus purpureus	Yeast, source of statins

(NCERT Ed. 2025-26, 12th, Page No. 153)

180. (1)

List-I	List-II
Rheumatoid arthritis	Autoimmune disorder
Tetanus	Antitoxin
Carcinogen	Neoplastic transformation
Malignant tumor	Metastatis

(NCERT Ed. 2025-26, 12th, Page No. 137)